MULTIPLE GLACIER SURGES OBSERVED WITH AIRBORNE AND SPACEBORNE INTERFEROMETRIC SYNTHETIC APERTURE RADAR

Brent Minchew[1], *Mark Simons*[1], *Scott Hensley*[2], *Helgi Björnsson*[3], *Finnur Pálsson*[3], *and Pietro Milillo*[4]

[1] Seismological Laboratory, California Institute of Technology, Pasadena, CA
[2] Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA
[3] Institute of Earth Sciences, University of Iceland, Reykjaík, Iceland
[4] School of Engineering, University of Basilicata, Potenza, Italy

1. INTRODUCTION

Mechanical properties of glacier beds impose fundamental constraints on glacier flow across a wide range of timescales [1]. Despite their importance in governing glacier dynamics, basal mechanics are not well understood, particularly where glaciers are underlain by deformable till [2]. While some till samples have been retrieved from beneath several glaciers and tested in laboratories in order to ascertain till rheology [3, 4], limitations on clast sizes imposed by apparatus dimensions and the difficulty of understanding and reproducing subglacial environments in the lab necessitate observations of the mechanical properties of *in situ* tills [5]. Such observations are sparse, owing to the inherent difficulty in attaining them, and this observational paucity has helped foment persistent uncertainties concerning the proper rheology of subglacial till and the rheological dependence on mechanical, thermal, and hydrological forcing [1, 2].

Observable transient behaviors offer key insights into the fundamental physical properties of natural systems. In the case of glaciers, we can glean insight into the rheology of subglacial tills by observing transient surges in glaciers known to be underlain by deformable tills. Surges, which are generally defined as relatively short timescale (months to years) episodes of rapid glacier flow interspersed in longer timescale (decades) quiescent phases, are facilitated through changes in the resistive shear traction at the ice-bed interface [6]. Where subglacial till is present, surges may arise from instabilities in the deformable substrate [7, 8]. By their transient nature, these instabilities help constrain the set of admissible models of till rheology; only models that can account for unstable deformation are viable.

Here we use synoptic-scale, observations of the surface velocity fields of Hofsjökull, a temperate Iceland ice cap known to be underlain by plastically deforming sediment [9, 10, 11], to study till rheology. Repeat-pass interferometric synthetic aperture radar (InSAR) observations were collected by European Remote Sensing 1 (ERS-1) in February 1994 [12], the middle of winter, and by the Uninhabited Aerial Vehicle Synthetic Aperture Radar (UAVSAR) in June 2012, beginning approximately 2 weeks after the onset of seasonal melt, and February 2014 [13]. By comparing surface speeds observed with UAVSAR and ERS, we note that wintertime speeds in 1994 were significantly higher than in winter 2014 and summer 2012 in three outlet glaciers (Fig. 1). Changes in glacier geometry in the interim between observations cannot account for the higher speeds in 1994, supporting conclusions from previous studies that these three outlet glaciers were surging in 1994 [13]. To study the mechanical properties of the till, we infer the basal shear traction using a numerical model constrained with the surface velocity observations to show that basal shear traction is lower during the surges, indicating the possibility of a rate-weakening subglacial till rheology.

2. DATA ACQUISITION AND PROCESSING METHODOLOGY

We follow approximately the same InSAR processing strategy for both the ERS and UAVSAR line-of-sight (LOS) data. All of the InSAR processing uses a lidar-derived DEM, which combines data collected in 2010 ($\sim 80\%$ coverage) and 2012 over Hofsjökull [14], blended with an ASTER (version 1) DEM for the surrounding areas. In all cases, the processed data provide at least three unique observational vectors, a fact we exploit in order to estimate the 3D velocity fields using a Bayesian approach [13]. This approach employs a spatially varying damping factor that reduces high-frequency variations in the final data and facilitates a relatively smooth transition across InSAR scene boundaries. Due to the lower signal-to-noise ratio in the vertical velocity component, which is primarily attributable to lower vertical speed relative to the horizontal speed, we utilize only the horizontal velocity components in this study, capitalizing on serendipitous observations of active glacier surges and quiescent periods afforded by InSAR collected at periods comparable to the length of typical surge cycles on Hofsjökull [9].

ERS-1 was a C-band (5.6-cm wavelength; 5.3 GHz),

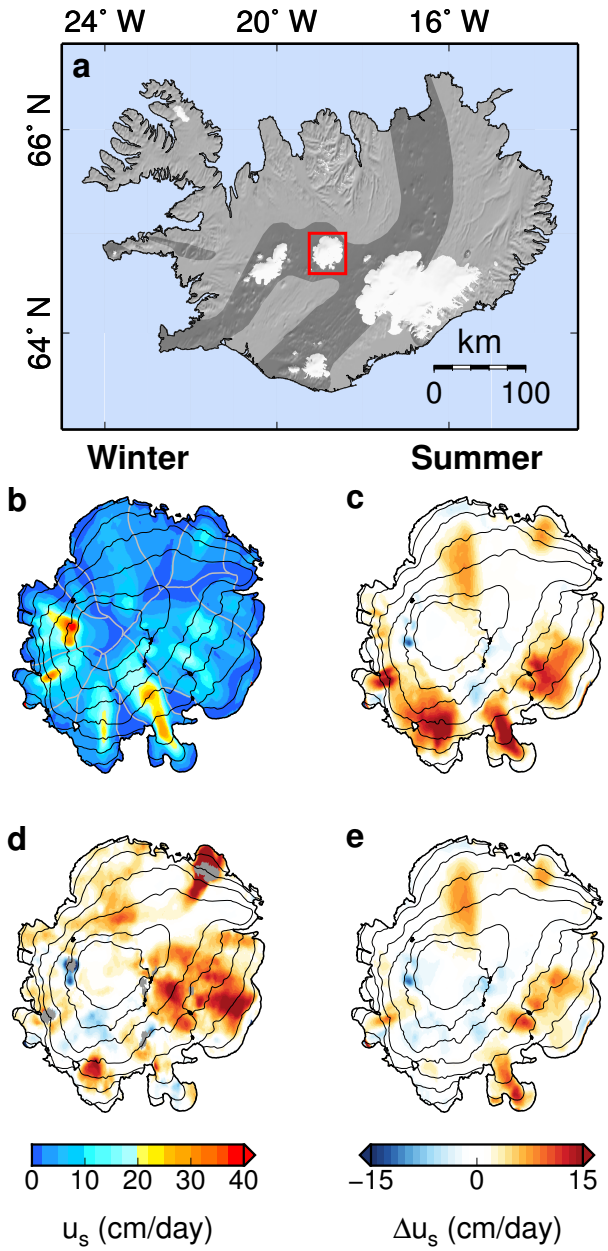

Fig. 1. **a** Shaded relief map of Iceland. Glaciers are white, Hofsjökull is enclosed by the red box, and darkened regions delineate volcanic zones. **b–e** Observed horizontal surface speeds and differential surface speeds. Data in the left and right columns were collected in winter and summer, respectively. **b** Horizontal surface speed inferred from data collected February 1–3, 2014 with UAVSAR. **c–e** Horizontal surface speed relative to **b** inferred from data collected **c** June 3–4, 2012 with UAVSAR, **d** February 1994 with ERS, and **e** June 13–14, 2012 with UAVSAR. Contour lines in **b–e** denote ice surface elevation in 150-m increments; maximum contour line is at 1650 m AMSL.

spaceborne SAR platform capable of collecting data with 30-m spatial resolution. Here we use data acquired on February 22 and 25, 1994 from a descending orbit and February 24 and 27, 1994 from an ascending orbit. We used the Repeat Orbit Interferometry Package (ROI_PAC) [15] to derive conventional, radar line-of-sight, interferograms and applied multi-aperture interferometry (MAI) [16] to estimate offsets along the satellite velocity vector for the ascending orbit. We do not use MAI results from the descending orbit due to unacceptably high noise. Applying the two interferograms and one MAI image, we inferred the 3D velocity field for all of Hofsjökull. We masked small areas of unacceptably high noise (gray patches in Fig. 1d), likely caused by rapid motion, and interpolated over the holes in order to initialize the ice flow models discussed in the next section. We applied zero weighting to those areas when inverting for basal mechanics (see next section).

UAVSAR is an L-band (24-cm wavelength; 1.25 GHz), airborne SAR platform capable of collecting data with raw resolution finer than 2 m. On June 3 and 4, 2012, June 13 and 14, 2012, and February 1 and 3, 2014, UAVSAR collected data over Hofsjökull from NASA's Gulfstream III aircraft along 9 flight tracks. The programability of UAVSAR flight lines allowed us to observe all of Hofsjökull from at least three unique LOS directions that have approximately equal azimuthal spacing. This coverage allows us to use only LOS InSAR data to infer the velocity fields, negating the need for noisier along-track displacement estimates available from MAI or 2D cross correlation methods [17]. We processed all of the raw data with the custom UAVSAR processing suite and incorporated data collected on subsequent days to form multiple 3D velocity fields for different data acquisition times [13]. The resulting velocity fields have formal errors less than 1 cm/day [11, 13].

Comparing data collected in February 1994 and February 2014 reveals that three outlet glaciers were moving markedly faster in 1994. These glaciers are Illviðrajökull in the northeast, Þjórsárjökull in the east, and Blautukvislarjökull in the south (Fig. 1b and d). Spatial patterns and amplitudes of these fast-moving glaciers are inconsistent with seasonal speedups (Fig. 1c and e), which are driven by surface melt. In 1994, Illviðrajökull was moving more than twice its summertime 2012 speed and rapid flow extended along the length of the glacier in 1994, as opposed to being more focused on the midstream areas as in 2012. Þjórsárjökull also experienced higher flow speeds in winter 1994 as compared to the early 2012 melt season. Notably, areas of enhanced flow are evident near the glacier terminus whereas areas of rapid flow caused by seasonal melt tend to be centered at approximately 1050 m elevation. Fast flow on Blautukvislarjökull during the surge is concentrated slightly west of the centroid of faster flow during melt season and at approximately the same elevation. During winter, air temperatures are well below freezing over the entire ice cap and no volcanic activity is known to

have occurred on Hofsjökull in 1994, negating the possibility of enhanced water flux at the bed being the driver of observed rapid flow in the three anomalously fast glaciers. Based on these observations, we are confident that these three outlet glaciers were surging during February 1994 [9, 13]. While it remains unknown when these surges began and ended, we can glean some insight into the properties of subglacial till by comparing surge and quiescent phase observations.

3. INFERRED MECHANICAL BED PROPERTIES

To ascertain the mechanical properties of the bed during the observational periods, we use the Ice Sheet Systems Model (ISSM) [18], a finite-element software suite, to infer the basal shear traction, τ_b, using the observed velocity fields as constraints on the ice flow. We construct the model domain using the DEM discussed above and a basal topography map derived from ice-penetrating radar surveys [19]. Our anisotropic mesh has a final resolution between 100 m and 500 m, with finer resolution concentrated in fast-flowing areas of the ice cap. We employ shallow-shelf assumptions for ice flow [1], assume a power-law basal boundary condition that relates basal shear traction to basal slip rate, u_b, ($\tau_b = C u_b^{1/m}$, where $C \geq 0$ and $m = 5$), and solve for C such that τ_b satisfies global stress balance and u_b minimizes the misfit between observed and modeled surface velocities [18]. Hofsjökull is composed of temperate ice and we chose the ice rheological parameters accordingly [1]. We assume that the rheological properties of ice are spatially and temporally constant, both reasonable assumptions given the timescales, spatial scales, and temperate nature of the ice.

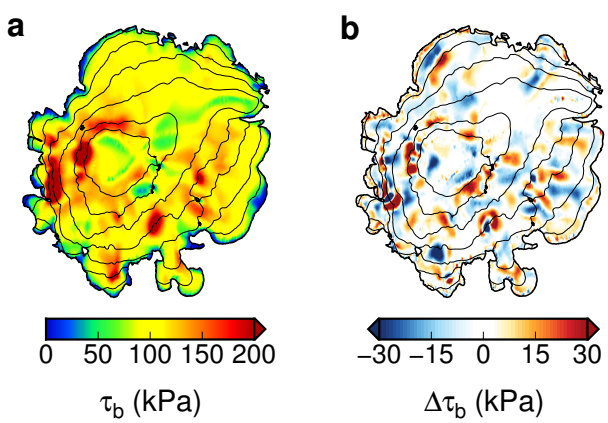

Fig. 2. **a** Inferred basal shear traction during the quiescent phase, February 2014. **b** Inferred basal shear traction during the surge (February 1994) phase relative to the quiescent phase. Negative (positive) values indicate lower (higher) stresses during the surge. Contour lines are the same as in Fig. 1.

In the three surging glaciers, inferred basal shear tractions are lower during surges than during the quiescent phases (Fig. 2). During February 2014, a quiescent period, areas that exhibit reduced surge-phase basal shear traction also showed little or no basal slip [11]. Comparing the surge speedup to basal weakening we note that reductions of basal shear traction of order 30 percent lead to speedups that more than double quiescent wintertime speeds. Nonlinearity between changes in basal shear traction and ice flow speed is a consequence of the non-Newtonian rheology of ice [11].

4. DISCUSSION

Lack of significant basal slip facilitates compaction of the underlying till, which can cause a transient increase in basal shear stress with the onset of basal slip through dilatant hardening [2, 4]. If the quiescent flow speed is less than the balance velocity of the glacier, which is true for all known surge-type glaciers in Iceland [9], this transient strain hardening, will allow for thickening and steepening of the ice that can effectively overload the glacier to a state that the gravitational driving stress exceeds the ultimate yield stress of the bed without inducing basal slip [4]. Thickening of the ice and steepening of the ice surface both increase gravitational driving stress, which scales as the product of ice thickness and ice surface slope. When the driving stress exceeds the peak strength of the bed, the glacier will slip as the bed deforms. Once slip has commenced, there will be a transient decrease in bed strength, which will approach steady-state as strain continues [2, 3]. Because the rate of change in bed strength will be much greater than the rate of change in gravitational driving stress, transient weakening of the bed reduces the ratio of basal shear traction to gravitational driving stress thereby increasing basal slip rate [11, 20].

The need for compaction of the till to facilitate transient increases in bed strength is an important feature in connecting observed glacier surge behavior with mechanical models of till. Studies of soil mechanics have shown that underconsolidated soils tend to compact when sheared while compacted soils dilate [21]. Dilation increases the porosity of till, which strengthens because of reduced pore water pressure [4]. Porosity can not increase indefinitely and must tend toward a steady state, thereby helping to set the peak bed strength. Once peak strength is exceeded, data from soil and fault-gauge experiments show that there is a transient weakening of the bed with shear [21, 4]. Overloading of gravitational driving stress and strain weakening provide valuable insight into the mechanism of incipient surge motion, a topic we will explore in a future paper.

The most significant limitations of this analysis are the lack of contemporaneous DEMs, which prevents us from properly constraining gravitational driving stress during each InSAR observation, and the lack of knowledge of when the observed surges began and ended. SAR platforms like

UAVSAR provide extensive programability for observational coverage and frequency, which can be leveraged to provide detailed, continuous coverage of smaller glaciated areas that may be difficult to observe with satellites, for a variety of reasons. Through approaches similar to the one described in this study, data from continuous, targeted observations of a few surge-type glaciers will help to constrain mechanical models of glacial till, leading to a more profound understanding of one of the most important open questions in glaciology: the mechanics of deformable glacier beds.

5. REFERENCES

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